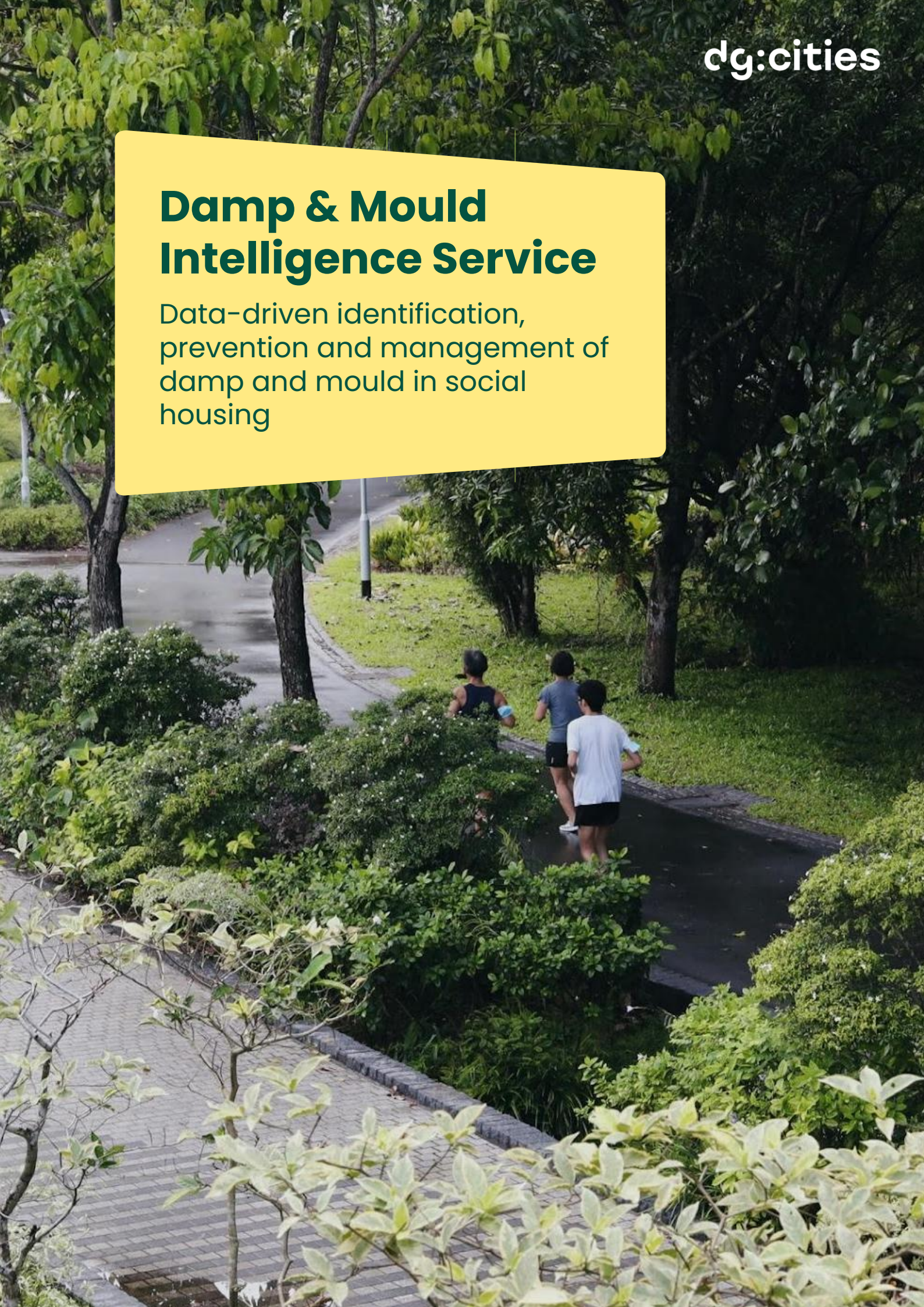
dg:cities

Damp & Mould Intelligence Service

Data-driven identification, prevention and management of damp and mould in social housing



About DG Cities

We help public sector clients harness the potential of data and insights with a practical approach that understands impact and puts people first.

DG Cities was established by the Royal Borough of Greenwich in 2015 to advance innovation in cities. We specialise in data-driven asset management, housing innovation, retrofit and decarbonisation, IoT deployment, business case development and impact evaluation.

Within the housing sector, our work is at the intersection of housing operations, asset strategy, compliance, technology providers and data integration platforms, helping landlords move from reactive approaches to intelligence-led decision making.

Why Damp & Mould Intelligence Matters

Damp and mould remain one of the most significant operational, financial and compliance challenges facing social landlords. Regulatory pressure, increasing reactive repair costs and limited visibility between inspections mean traditional approaches are often reactive and resource intensive.

Environmental monitoring enables landlords to move toward a proactive model — identifying risk earlier, targeting interventions more effectively, and linking damp risk to broader asset and retrofit strategies.

What we deliver

A critical friend to project teams

DG Cities supports landlords to move from reactive damp and mould management to a proactive, intelligence-led approach. By combining environmental monitoring, housing data integration and financial modelling, we transform sensor data into operational insight and strategic decision making. Our approach strengthens compliance, improves targeting of interventions and supports long term asset performance across housing portfolios.

We use quantitative and qualitative approaches to create a rich understanding of a project or approach, and work with clients to explore what works, what doesn't and ways to improve.

We help clients understand what works, where and why, through a variety of methods and approaches, including:

- **Sensor Deployment & Strategy:** We help landlords deploy environmental sensors where it will deliver the greatest impact, saving on overall programme costs. Using asset data, archetypes and known risk indicators, we identify priority homes and align deployment with retrofit and decarbonisation programmes. All outputs are structured to support compliance, governance and clear executive reporting.
- **Data Integration & Dashboarding:** Sensor data becomes truly valuable when connected to other existing data. We integrate environmental monitoring with repairs data, EPC ratings and property archetypes to create a unified view of risk. Our prioritisation tool provides clear property level and portfolio level visibility, enabling teams to prioritise interventions with confidence, using a platform like PowerBI for a recognisable user interface.
- **Insights & analysis:** We turn environmental data into actionable insight. By identifying high risk homes early and analysing links between reactive spend and environmental conditions, we shift focus to proactive interventions. We also track the impact of works over time, helping organisations demonstrate measurable improvement to regulators, leadership and residents.
- **Business Case & Financial Modelling:** We support strategic decision making through robust financial modelling. This includes long-term lifecycle analysis, scenario modelling and comparisons between reactive and sensor enabled approaches. Our modelling quantifies avoided costs, operational efficiencies and return on investment, supporting confident scale up decisions.
- **Process Transformation:** Our Damp & Mould Intelligence Framework embeds monitoring into everyday operations - from identifying portfolio risk, to targeted deployment, data integration and operational response. Continuous review ensures insights translate into improved performance, reduced risk and stronger asset strategy over time.

Case Study: Royal Borough of Greenwich

DG Cities partnered with the Royal Borough of Greenwich to design and deliver a data led damp and mould monitoring pilot across the housing portfolio. The objective was to move from a reactive, inspection led model toward proactive risk identification using environmental sensors integrated with housing data systems.

Approximately 160 environmental sensors were deployed across selected homes, with monitoring undertaken over a 24-month period. Sensor data was integrated with repairs records, property archetype data and retrofit programme information to create a holistic view of damp and mould risk.

"From the outset DG Cities demonstrated a thorough understanding of our objectives and tailored their approach to meet our needs. They delivered comprehensive insights that have been invaluable to us."

Ryal Borough of Greenwich, Damp, Mould & Condensation Manager

Measuring the Impact of Winter Interventions

Seasonal risk is a critical pressure point for damp and mould. Using sensor data, we analysed winter mould risk scores and assessed how targeted interventions influenced outcomes during peak risk periods.

How effective are different interventions at reducing damp risk?

Category

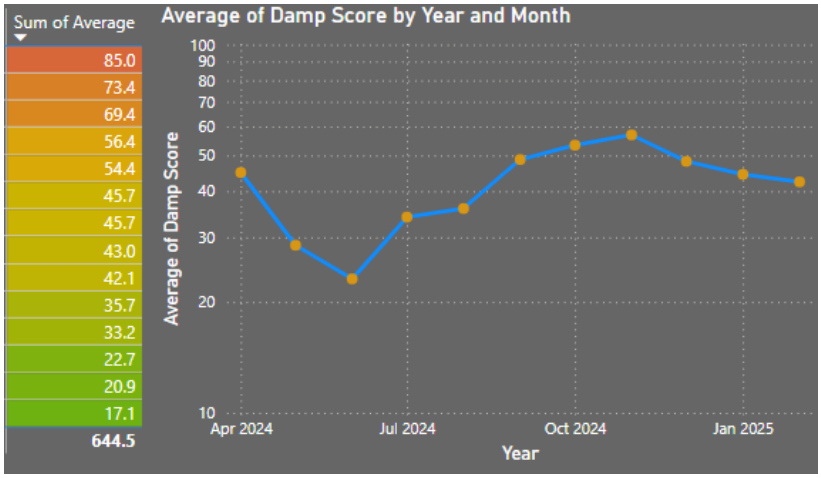
- Bath Panel
- Blockage
- Ceiling
- Damp Works
- Door
- Extractor Fan
- Guttering
- Leak
- Loft Insulation
- Mould Wash
- New Kitchen

Category	Count	Change in Winter Scores
Roof Repairs	22	-11.56
Repoint	6	-11.30
Ventilation	4	-10.85
Shower	16	-10.49
Plumbing	37	-4.52
Thermal Bo...	16	-4.40
Ceiling	28	-1.40
Window	75	-0.59
Pacifier Vent	17	-0.54
Other	100	-0.53
Mould Wash	161	-0.31
Plastering	10	-0.19
Bath Panel	27	-0.17
Guttering	11	0.16
Leak	164	0.39
Total	**945**	**0.87**

The analysis compared properties before and after specific interventions, highlighting measurable reductions in winter risk scores where roof repairs, repointing or ventilation works had been completed. This enabled the Council to move beyond anecdotal feedback and objectively assess whether interventions were delivering tangible performance improvements.

Evaluating Whole-House Retrofit Performance

Environmental monitoring also provided an opportunity to assess the wider impact of retrofit measures delivered through whole house improvement programmes e.g. SHDF.



By tracking damp and mould risk scores in homes that had undergone whole-house retrofit works, we were able to evaluate how fabric and ventilation upgrades influenced environmental conditions. The analysis demonstrated where retrofit programmes were contributing to reduced mould risk - and where additional behavioural or ventilation measures may still be required. This insight supports alignment between damp management and decarbonisation strategy.

Identifying Asset-Level Risk Patterns

To understand whether high-risk readings were isolated anomalies or indicative of wider building issues, we compared sensor scores against asset group data - including the proportion of homes within the same asset group with historic damp or leak-related repairs.

Damp Sensor Score	% of asset with damp repair history	% of asset with leak repair history
91.84	66.7%	75.76%
68.34	61.8%	64.71%
70.61	60.0%	77.14%
91.97	50.0%	65.38%
29.06	50.0%	65.38%
48.00	46.4%	67.86%
98.43	46.4%	67.86%
72.51	45.2%	64.29%
61.30	45.2%	64.29%
92.79	45.2%	64.29%

This comparison enabled us to determine whether elevated risk was likely property specific or potentially systemic within a block or asset group. The approach strengthened asset intelligence, allowing teams to prioritise investigations and capital planning more strategically.

Demonstrating Proactive Repair Savings

One of the most significant findings was the ability to evidence proactive intervention. We compared properties within the same block, distinguishing between cases where sensors triggered early repair activity and those where action only followed a resident report.

Property 1:

Property	Issue	Cause	Repair	Cost	Outcome
Property One - Sensor Used	Leak detected early via sensor alert	Leak from upstairs bathroom into living room & hallway	Bathroom & kitchen revive fans; patch leak in living room; passive vent to toilet; renew hallway	£3,094	Early intervention limited damage; resolved through planned repair
Property Two - Non Sensor Used	Leak reported by Occupational Health Team	Identical leak from upstairs bathroom into living room & hallway	Bathroom and hallway repairs; extensive redecoration due to prolonged exposure	£6,577	Late identification led to greater damage and higher repair cost

Property 2:

Property	Issue	Cause	Repair	Cost	Outcome
Property Three - Sensor Used	High humidity levels in bathroom	Faulty fan which had stopped working	New Fan installed	£448	Early intervention limited damage
Property Four - Non Sensor Used	Damp/Mould within bathroom, extensive damage	Faulty fan led to condensation which damamged plaster/tiles	renewal of plaster, tiling, new fan install	£1,223	Late identification led to greater damage and higher repair cost

The results highlighted how early sensor-led identification reduced escalation, supported earlier repair resolution and demonstrated potential operational savings. This provided a clear financial and service improvement narrative to support wider deployment.

Scalable Intelligence Framework

The Greenwich pilot demonstrated that environmental monitoring, when integrated with housing data systems, can:

- Identify risk before complaint
- Quantify intervention effectiveness
- Strengthen asset-level intelligence
- Align damp strategy with retrofit programmes
- Support a robust financial case for scale

The approach established a scalable framework that embeds environmental data into operational decision-making, compliance reporting and long-term asset planning.

Our team

Balazs Csuvar
Director of Innovation and Net Zero
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Balazs leads the delivery of our innovation projects, solving challenges through the integration of new technologies and holistic thinking. He is a civil engineer by background, with 10 years of international experience as an urban consultant, leading a wide range of projects from developing smart city strategies, planning connected and autonomous mobility, and in local government capacity building. His current focus is on harnessing AI for data analysis and to rethink public sector service delivery.

Rasheed Sokunbi
Innovation and Net Zero Consultant
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As a specialist in Net Zero, Rasheed has been integral in developing our work on sensors in Greenwich and beyond. Previously, he worked for one of UK's largest housing associations as an energy advisor, where he gained exposure to district and communal heating systems, and for a sustainable energy consulting firm, managing retrofit meter exchange projects.

Nima Karshenas
Innovation and Net Zero Consultant
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A MEng graduate in Electrical and Electronic Engineering from Imperial College London, Nima contributes his in-depth technical knowledge of AI and statistics to diverse projects that promote social impact and sustainability. His work focuses on delivering AI implementation as part of the Home-by-home plan tool, linking sensor data with wider housing datasets.

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Thank you

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